

Solar Day Lamp With Battery Backup

Dr Vijay Deshpande

This solar day lamp with battery backup can be used in offices, homes, shops, warehouses, temples, etc. It offers a cost-effective way of harnessing solar energy. It is simple to manufacture and has a long life.



Fig. 1: Author's prototype of solar day lamp with battery backup

The basic solar day lamp (SDL) consists of only two parts: photovoltaic (PV) panels and LED arrays. Due to simple construction, the SDL offers most cost-effective way of harnessing solar energy (off grid). SDL is ideally suited for offices working from 9am to 5pm. It provides light throughout the day.

However, this simple design has some drawbacks. It leads to energy wastage during weekends when the office is closed. Also, energy is wasted from sunrise till the time the office opens.

In order to improve the energy utilisation of SDL, a partial battery backup system is proposed here. The

Box 1: PV panel and battery specifications

PV power rating (W_p) = $2 \times 10 = 20W_p$ (Watts peak)
 Current at maximum power (I_{mp}) = 0.58A
 Energy generated by panels in 8 hrs = $35 \times 0.58 \times 8 = 162.4Wh$
 Total energy generated per week = 1136.8Wh
 Storage capacity of battery = 7Ah
 Battery bank voltage when fully charged (V_{bmax}) = $13 \times 3 = 39V$
 Battery bank voltage when fully discharged (V_{bmin}) = $11.5 \times 3 = 34.5V$
 Average energy stored in the battery = $7 \times (39 + 34.5)/2 = 257.25Wh$
 Energy stored in the battery/week = $(257.25/1136.8) \times 100 = 22.5\%$
 Power rating of white LED = 1W
 Forward voltage of LED = 3V
 LED current at full power (I_{LEDmax}) = $1/3 = 0.33A$

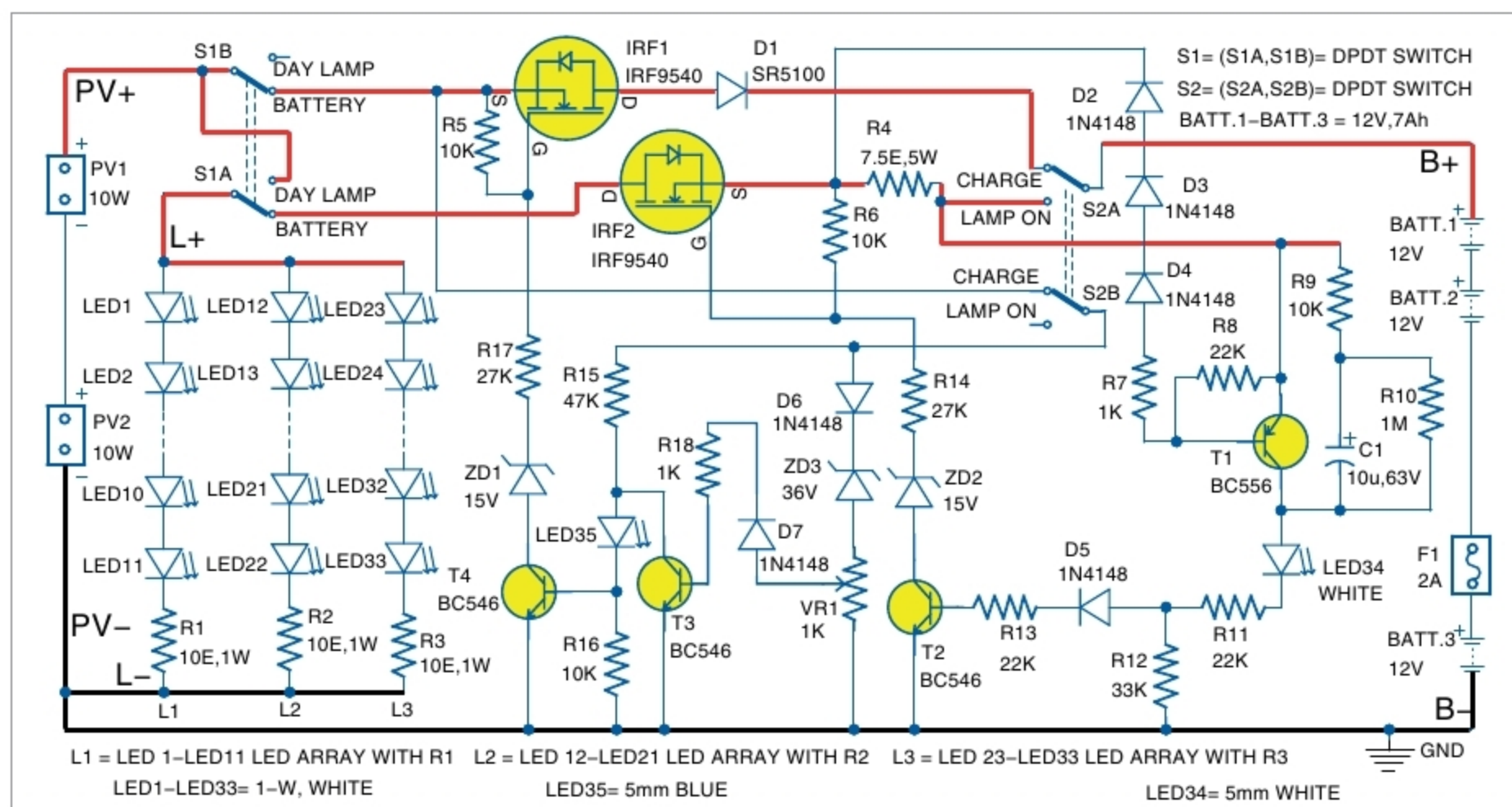


Fig. 2: Circuit diagram of solar day lamp with battery backup